



Inactivation of Conserved *C. elegans* Genes Engages Pathogen- and Xenobiotic-Associated Defenses

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SUMMARY

The nematode *C. elegans* is attracted to nutritious bacteria and is repelled by pathogens and toxins. Here we show that RNAi and toxin-mediated disruption of core cellular activities, including translation, respiration, and protein turnover, stimulate behavioral avoidance of normally attractive bacteria. RNAi of these and other essential processes induces expression of detoxification and innate immune effectors, even in the absence of toxins or pathogens. Disruption of core processes in non-neuronal tissues was sufficient to stimulate aversion behavior, revealing a neuroendocrine axis of control that additionally required serotonergic and Jnk kinase signaling pathways. We propose that surveillance pathways overseeing core cellular activities allow animals to detect invading pathogens that deploy toxins and virulence factors to undermine vital host functions. Variation in cellular surveillance and endocrine pathways controlling behavior, detoxification, and immunity selected by past toxin or microbial interactions could underlie aberrant responses to foods, medicines, and microbes.

INTRODUCTION

Organisms compete for finite resources, driving the evolution of interspecies conflict. Host-pathogen relationships are an example of this conflict, where the volley of measures and countermeasures is complex and highly evolved. For example, microbial pathogens release an array of toxins or virulence factors aimed at crippling host functions and defenses. To counter pathogen or toxic attack, the host mobilizes immunity and detoxification responses. Chemical and protein toxins are wildly diverse and operate by sabotaging critical cellular activities such as transcription, translation, or mitochondrial respiration. If the host is to survive such an assault, it must first detect the attack, then mobilize xenobiotic detoxification programs, compensatory mechanisms to rescue disabled cellular processes, and immu-

nity-mediated eradication of the toxin's microbial source. A major question in innate immunity is how hosts detect pathogens and xenobiotics. One possibility, called the PAMP (pathogen-associated molecular pattern) hypothesis, is that cellular receptors (such as nuclear hormone receptors or G protein-coupled receptors [NHRs or GPCRs] or leucine-rich repeat [LRR] proteins) recognize specific toxins or microbial products (Janeway, 1989; Jones and Dangl, 2006). Such a solution is limited by the chemical specificity of cellular receptors (including receptors with chemical promiscuity) and may not protect the host from pathogens with which it has no evolutionary history. A more adaptable solution, called the DAMP (damage-associated molecular pattern) hypothesis, involves molecular recognition of damaged extracellular matrix components or intracellular proteins and small molecules released into the extracellular environment following pathogen-induced cell lysis (Matzinger, 1994).

C. elegans is a motile, soil-dwelling metazoan with astute sensory capabilities that allow it to escape a broad range of environmental hazards, including pathogens (Schulenburg and Ewbank, 2007; Pujol et al., 2001; Zhang et al., 2005), although the mechanisms of pathogen sensation are poorly understood. Here we show that surveillance pathways in *C. elegans* monitor core cellular activities and interpret disruption of these activities as evidence of pathogen attack, engaging behavioral, immune, and detoxification responses. These cellular surveillance-activated detoxification and defenses (which we call cSADDs) monitor cellular components such as the ribosome, proteasome, and mitochondria, initiating aversion behavior and detoxification and innate immune defenses when cellular perturbations are detected. Surveillance of cellular machinery provides a highly generalizable mechanism for sensation of invading microbes and xenobiotic compounds.

RESULTS

Disruption of Core Cellular Processes by RNAi Stimulates Microbial Aversion Behavior

E. coli is a nutritive laboratory food source that is normally attractive to *C. elegans*. RNA interference (RNAi) is used to inactivate genes by feeding animals *E. coli*-expressing double-stranded RNAs (dsRNAs) complementary to individual *C. elegans* transcripts. RNAi libraries allow the inactivation of ~90% of the

C. elegans genome (Kamath et al., 2003). We tested whether RNAi of a small set of *C. elegans* genes could alter the affinity of animals for an *E. coli* lawn and found that inactivation of many genes essential for normal growth and development stimulated a behavioral avoidance phenotype (Figures 1A and 1B; Figures S1A and S1B and Movies S1, S2, and S3 available online), a response we refer to as “microbial aversion behavior.” The aversion phenotype could reflect a programmed behavioral response to inactivation of essential processes or simply a reduction in the nutritive quality of dsRNA-expressing *E. coli*. To distinguish these possibilities, we tested whether RNAi of these essential genes produced the aversion phenotype in RNAi-defective animals. The RNAi-defective *C. elegans* mutants *sid-1(qt9)* and *rde-1(ne219)* failed to exhibit the aversion response when grown on RNAi bacteria repulsive to wild-type controls (Figure S1A). In the absence of dsRNA targeting essential *C. elegans* pathways, basal frequencies of food avoidance were very low. For animals raised on RNAi control bacteria, aversion was undetectable ($0\% \pm 0\%$) in the first three larval stages (L1, L2, L3), became weakly detectable in the L4 stage ($0.5\% \pm 0.5\%$), and increased to $\sim 5\%$ in adults ($\pm 3\%$) (Figure S1C).

To identify disruptions capable of evoking an aversion response, we inactivated 4,062 essential and metabolic genes. Essential genes were defined as those required for first- or second-generation viability in whole-genome screens of wild-type *C. elegans* (N2) or enhanced RNAi strains. Metabolic genes were selected based on assembled Gene Ontology (GO) term annotations. We found that 379 of the 4,062 gene inactivations assayed stimulated bacterial aversion $>10\%$ (Table S1A). Many of the genes identified also caused larval delay or arrest phenotypes (Figures 1C, S1D, and S1E; Table S1A). The frequency of developmental phenotypes in the aversion gene set was significantly higher than in the set of 4,062 genes tested (85% versus 21% of genes, $p < .0001$, chi-square test) (Figures S1D and S1E). There was an ~ 10 -fold increased frequency of aversion phenotypes in the essential gene set compared to a set of 192 randomly selected genes ($p < .0001$, chi-square test) (data not shown).

DAVID gene enrichment analysis (Huang et al., 2009) revealed that inactivation of genes from particular functional classes preferentially induced the aversion response (Figure 1D). Enriched functional classes were comprised of genes involved in protein translation (25- and 27-fold enrichment of ribosome components and tRNA synthetases, respectively), the molting cycle (6-fold enriched), proteasome machinery (38-fold enriched), TCP chaperonins (25-fold enriched), mitochondria (7-fold enriched), vacuolar ATPases (18-fold enriched), mRNA processing (12-fold enriched), and histones (19-fold enriched). Although not statistically significant (possibly due to small sample size), cytoskeleton components were also enriched in the aversion gene set (~ 10 -fold and 20-fold enrichment for tubulin and actins, respectively). Moreover, we found that aversion-associated genes exhibited 2-fold higher conservation between *C. elegans* and *Homo sapiens* than the screen set and 4-fold higher conservation than the entire *C. elegans* genome, suggesting that ultraconserved cellular processes are more likely to be subject to a form of surveillance that controls adaptive behaviors (Table S1B).

Human homologs of many aversion genes have also been implicated in human disease (Table S1C).

Chemicals and Bacterial Toxins Stimulate Microbial Aversion

Many of the cellular processes identified by our RNAi screen are known targets of microbial toxins (Figure 2A). For example, the eukaryotic ribosome is inhibited by dozens of pathogen-derived toxins, including hygromycin, cycloheximide, and diphtheria toxins. The electron transport chain (ETC), the proteasome, tRNA synthetases, mRNA splicing, vacuolar ATPases, tubulins and actins were identified by our screen and are also targets of microbial toxins. Because wild *C. elegans* species live in microbe-rich environments, we hypothesized that RNAi of critical cellular pathways mimics the impact of microbial toxins on cellular physiology, and that surveillance pathways within the animal oversee cellular activities as a broadly adaptable means of toxin detection. Under these conditions, activation of an aversive behavioral program represents an adaptive response.

To test the hypothesis that aversion behavior is a generalized response to toxin exposure, we supplemented benign *E. coli* lawns with various poisons, many of which target essential activities identified in our screen, to determine whether animals developed aversion toward an otherwise attractive *E. coli* strain (OP50). Geneticin/G418 was used to inhibit translation, antimycin A to inhibit the mitochondrial ETC, bortezomib to inhibit proteasome activity, concanamycin A to inhibit vacuolar ATPase activity, tunicamycin to inhibit the secretory pathway, zeocin to induce genotoxic stress, and paraquat to induce oxidative stress. Notably, antimycin, tunicamycin, and concanamycin are toxins produced by *Streptomyces* species resident to soil and compost, both natural habitats for *C. elegans*. L4 larvae/young adults were placed on *E. coli* OP50 lawns supplemented with one of these compounds and assayed for aversion behavior over time. Exposure to any of these toxins stimulated avoidance of the microbial lawn, usually within 4–16 hr of exposure (Figures 2B and 2C). Treatment of larval stage animals produced similar results (data not shown). Aversion behaviors developed more rapidly in response to poisons than RNAi, probably because drugs directly inhibit protein activities, whereas RNAi degrades mRNAs to prevent new protein translation but does not inhibit pre-existing cellular proteins.

Inactivation of Essential Cellular Activities Stimulates Pathogen and Detoxification Responses

Because bacterial pathogens are known to stimulate food-avoidance behavior in *C. elegans*, the observation that inactivation of cellular processes using toxins or RNAi produced a similar behavioral response suggests an adaptable mechanism for toxin and pathogen detection mediated by cellular surveillance pathways in the host. We reasoned that if animals use internal physiologic cues to recognize the presence of pathogens, then perhaps RNAi of essential cellular components would trigger physiologic defenses associated with pathogen infection or toxin exposure—in the absence of pathogens or toxins.

To address whether aversion-inducing gene inactivations can engage transcriptional responses to pathogen exposure in *C. elegans*, we examined the expression of several GFP

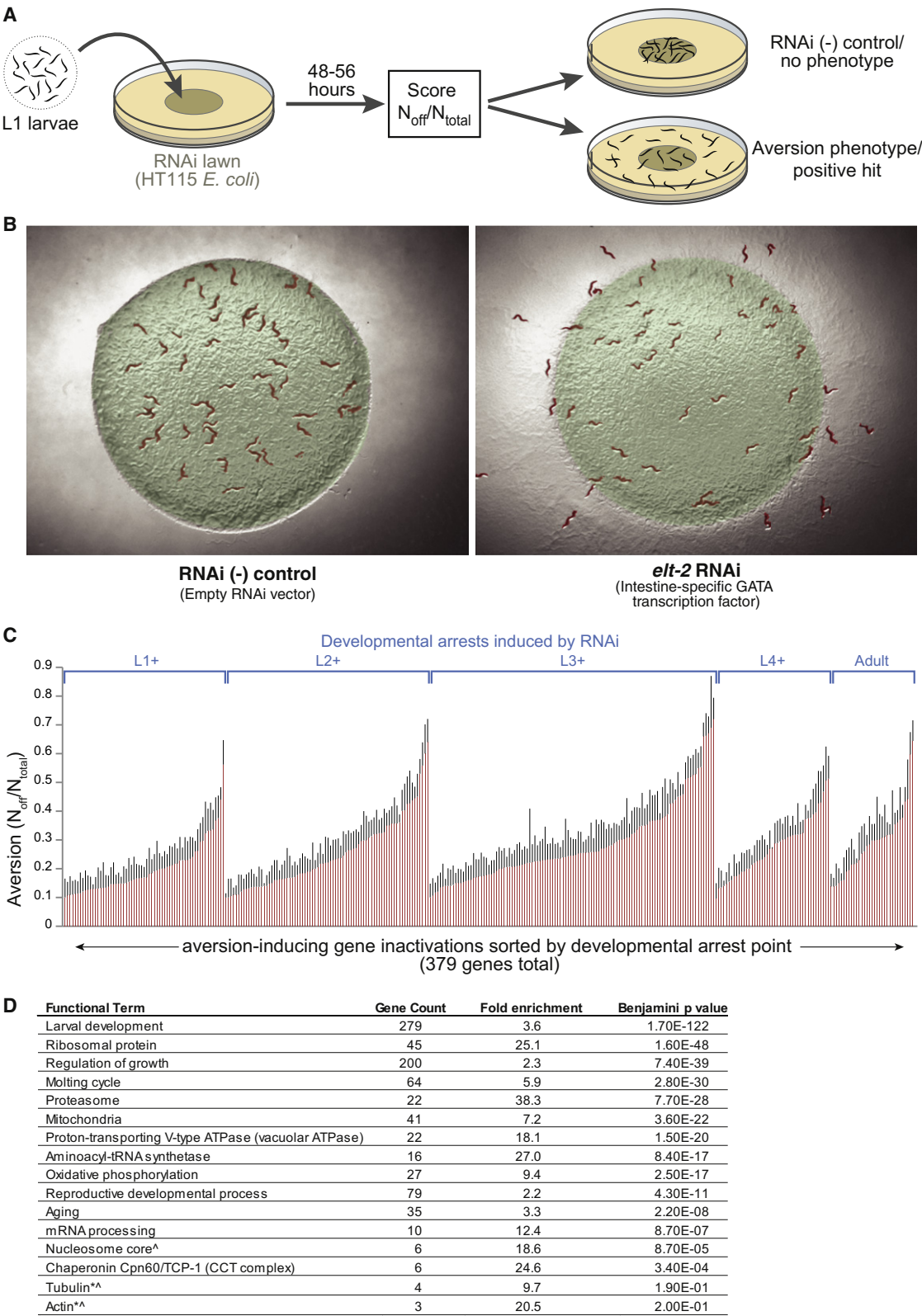


Figure 1. Inactivation of Essential Cellular Pathways Stimulates Microbial Avoidance Behavior
(A) Schematic of RNAi microbial aversion assay.
(B) Example of control and aversion phenotypes after 48 hr of growth on *elt-2* or RNAi control bacteria. Animal positions are highlighted in red.

reporters previously shown to represent activation of innate immune programs. For each reporter, GFP was fused to the promoter of a gene induced by a microbial pathogen but not by benign *E. coli* or attenuated pathogenic strains. Three of these genes have proposed functions in innate immunity because they contain conserved domains associated with innate immune responses in *C. elegans* and other species: *clec-60* (a C-type lectin/CUB domain protein induced by *S. aureus* and *M. nematophilum*) (O'Rourke et al., 2006), *F35E12.5* (a CUB domain protein induced by *Y. pestis*, *M. nematophilum*, and *P. aeruginosa*) (Troemel et al., 2006; O'Rourke et al., 2006; Bolz et al., 2010), and *nlp-29* (a conserved glycine/tyrosine-rich antimicrobial peptide induced by *D. coniospora*, *S. marcescens*, and wounding) (Pujol et al., 2008). Finally, *irg-1::GFP* is a reporter induced by *P. aeruginosa* and cadmium poisoning whose expression in response to pathogen exposure requires ZIP-2, a bZIP transcription factor (Estes et al., 2010). *clec-60::GFP*, *F35H12.5::GFP*, and *irg-1::GFP* are all primarily induced in the intestine, whereas *nlp-29::GFP* induction is generally hypodermal.

Induction of each pathogen reporter was analyzed against a panel of ~100 gene inactivations representing the major functional classes identified in our aversion screen and a random panel of genes from the whole-genome RNAi library. For all four pathogen-response genes, we observed a significantly elevated frequency of activation by the aversion gene set relative to the control set ($p < 0.0001$, chi-square test) (Figures 3 and S2C; Table S2A). The *clec-60::GFP* reporter was activated by 59% of aversion gene inactivations tested, *F35H12.5::GFP* by 24%, *irg-1::GFP* by 45%, and *nlp-29::GFP* by 28%, representing a 2- to 9-fold increase in induction frequencies relative to the random set (Figure S2C). In general, the three intestinal reporters were induced by RNAi against most functional classes, with most potent induction following disruption of protein synthesis and metabolic and mitochondrial functions (Table S2A). In contrast, the hypodermal *nlp-29::GFP* reporter was most consistently activated by disruption of the molting program. Starvation also induced the *clec-60::GFP* and *nlp-29::GFP* reporters.

If host surveillance pathways detect the presence of chemical toxins by monitoring core cellular activities, then RNAi-mediated disruption of these activities might also stimulate drug detoxification responses, such as cytochrome P450 and glutathione S-transferase genes. To test this hypothesis, we assayed GFP transcriptional reporters for induction of genes encoding xenobiotic detoxification enzymes, *cyp-35B1* (an intestinally expressed cytochrome P450 oxidase) (Iser et al., 2011) and *gst-4* (a glutathione S-transferase induced by drugs and toxins) (Figures 3B and 3C). Each of these reporters was induced at a frequency ~4-fold above background (~40% of aversion-inducing essential gene inactivations tested induced *cyp-35B1::GFP*, and ~20% induced *gst-4::GFP*, $p < 0.0001$, chi-square test) (Figure S2C). The pattern of gene inactivations that stimulated expression of xenobiotic detoxification responses was overlap-

ping but distinct (Table S2A). Both reporters were robustly induced by inactivation of genes involved in protein synthesis, mitochondrial function, and metabolism. *gst-4::GFP* alone was induced by RNAi of proteasome subunits, and *cyp-35B1::GFP* alone was induced by RNAi of splicing and vesicular trafficking components. The *cyp-35B1::GFP* reporter was also induced by inactivation of many molting and cuticle-specific genes, despite an expression pattern that was limited to the intestine.

General reporters of cellular stress, such as the mitochondrial unfolded protein response (UPR^{mito}) and the endoplasmic reticulum unfolded protein response (ER^{UPR}), assayed using *hsp-6::GFP* (Yoneda et al., 2004) and *hsp-4::GFP* (Calfon et al., 2002; Urano et al., 2002) were also triggered by many essential gene inactivations, as was a *sod-3::GFP* oxidative stress reporter (Libina et al., 2003) and a *gpdh-1::GFP* osmotic stress reporter (Lamitina et al., 2006). The majority of gene inactivations resulted in activation of at least one cellular stress reporter, and many gene inactivations stimulated more than one reporter (Figure S2; Table S2B).

Microbial Aversion Behavior Is Controlled by a Neuroendocrine Circuit

Neurosensory inputs are necessary for pathogen-avoidance behavior (Styer et al., 2008), but the contribution of sensory signals originating within an animal's internal tissues has been difficult to determine. Three lines of evidence suggest that aversion behavior induced by essential gene inactivations is not exclusively a neuronal response but is also mediated by endocrine signaling between the *C. elegans* nervous system and other tissues: (1) both drugs and RNAi targeting the same process can stimulate aversion, suggesting that drug detection by neuronal chemoreceptors is not essential; (2) RNAi works poorly in *C. elegans* neurons and almost always requires an enhanced RNAi mutant strain to elicit detectable phenotypes (Calixto et al., 2010); and (3) a subset of genes identified in the aversion screen exhibit postembryonic expression patterns that exclude the nervous system but include the intestine, hypodermis, gonad, pharynx, vulva, body wall muscle, and/or the excretory cell, suggesting that any or all of these tissues may be capable of triggering a neuroendocrine-mediated aversion response (Table S3). For example, the ELT-2 GATA transcription factor acts exclusively in the intestine to control gut development (Fukushige et al., 1998), and the QUA-1 hedgehog homolog acts in the hypodermis to control molting (Hao et al., 2006). Thus the aversion response to inactivation of essential genes is likely to operate through an endocrine-mediated relay of stress signals between non-neuronal tissues and the neurons that generate aversive behavior.

To test directly whether non-neuronal cues can trigger the aversion response, we generated *C. elegans* strains in which RNAi operates only in specified tissues by tissue-restricted restoration of RNAi in RNAi-defective mutants (either *rde-1(ne219)* or *sid-1(qt9)*). Tissue-specific rescue of *rde-1* or *sid-1*

(C) Developmental stage- and rank-ordered aversion levels + SEM for 379 gene inactivations (of 4,062 screened) exhibiting aversion $\geq 10\%$ off bacteria at 48–58 hr of growth on RNAi bacteria.

(D) DAVID bioinformatic analysis of aversion genes showing enrichment for specific functional categories. * denotes enriched gene classes that were not statistically significant due to small N. ^ denotes high intra-class homology that could produce an elevated false-positive rate due to off-target RNAi effects. See also Figure S1, Table S1, and Movies S1, S2, and S3.

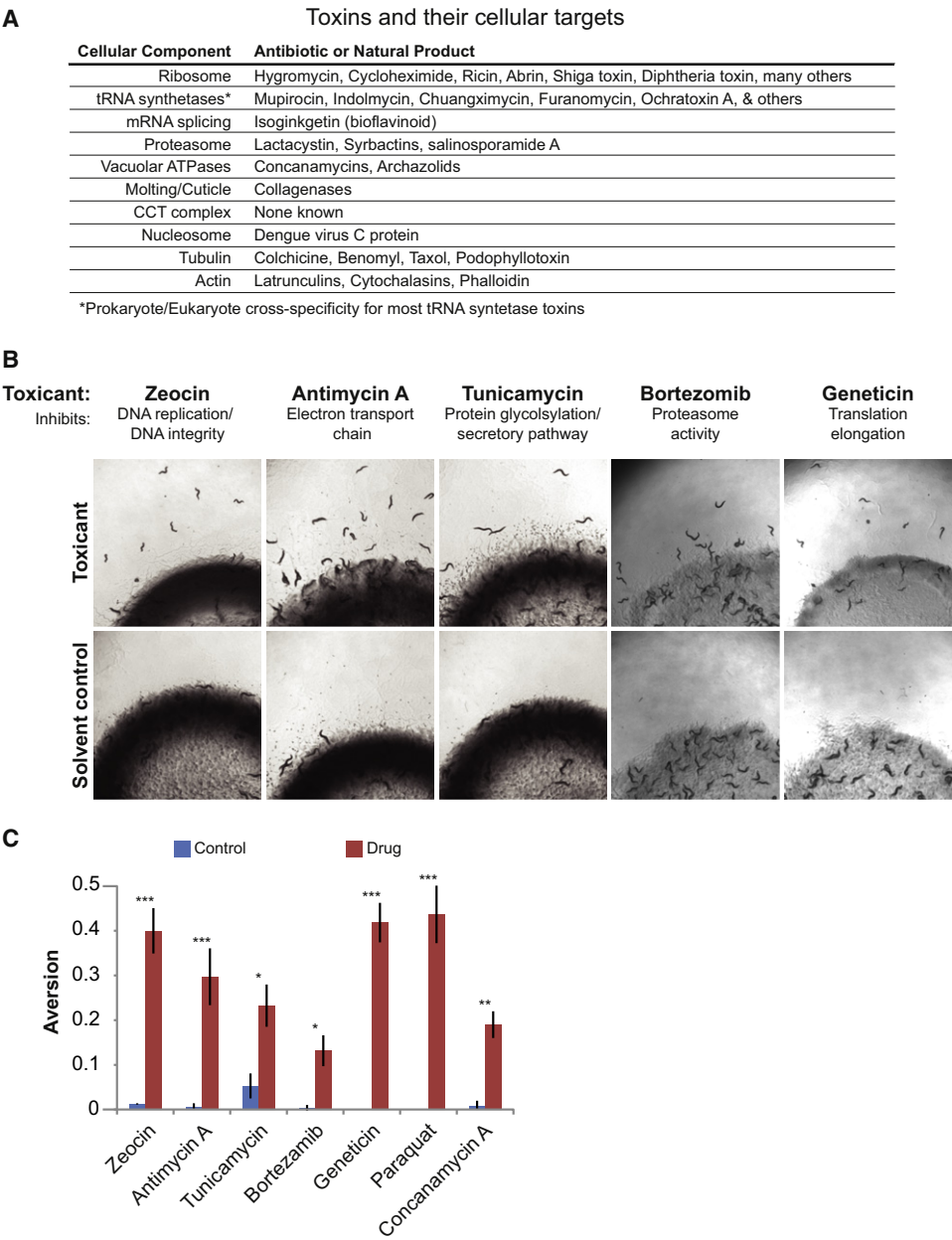


Figure 2. Toxins Stimulate the Microbial Aversion Behavior
(A) A partial list of microbially synthesized toxins that target the cellular processes identified in the microbial aversion RNAi screen.
(B) Toxin-induced microbial aversion phenotypes. Drug concentrations are provided in [Extended Experimental Procedures](#).
(C) Aversion levels $\pm$ SEM for each drug at peak time points: zeocin (16 hr); paraquat (8 hr); antimycin A (6 hr); tunicamycin (24 hr); geneticin (3 hr); bortezomib (6 hr); concanamycin A (8 hr). * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$ by t test.

null mutants can restrict RNAi-based gene inactivation to select tissues (Qadota et al., 2007; Jose et al., 2009). *rde-1* expression from the hypodermal *wrt-2* promoter was used to restore RNAi activity to hypodermal and seam cells in *rde-1(ne219)* animals. A gut-specific *vha-6* promoter driving expression of *sid-1* restored RNAi to intestinal cells in a *sid-1(qt9)* mutant. Tissue specificity for RNAi was confirmed by introducing a ubiquitously expressed GFP reporter (*sur-5::GFP^{NLS}*) into

hypodermal and intestinal RNAi lines and performing RNAi against *GFP* (Figures S3A–S3D). In *wrt-2::RDE-1* animals, GFP silencing was observed in the hypodermis but not in other major tissues (the muscle, intestine, neurons, or pharynx). In the *vha-6::SID-1* line, GFP was silenced in the intestine but not elsewhere. As further reassurance of the tissue specificity of these lines, we inactivated several genes with known restricted patterns of expression (Table S3) and found that aversion was

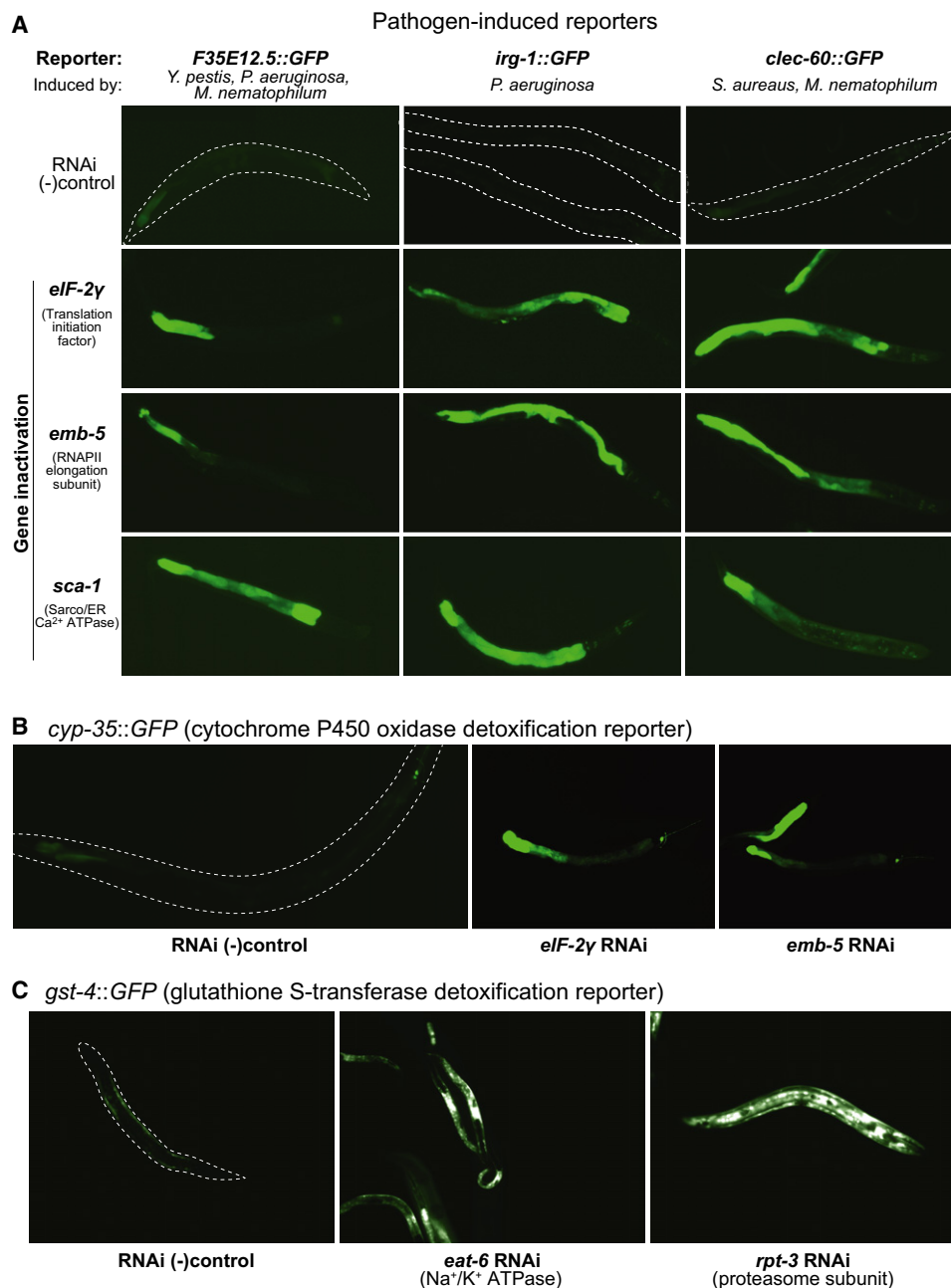


Figure 3. Activation of Pathogen-Associated and Detoxification Responses

(A) Innate immunity-associated reporters are induced in the intestine in response to growth on RNAi lawns inactivating the *C. elegans* *elF-2γ* homolog Y39G10AR.8, *emb-5*, and *sca-1* genes. Dashed lines outline the bodies of RNAi control-treated animals.

(B) RNAi of *elF-2γ* and *emb-5* stimulates intestinal expression of a *cyp-35B1::GFP* transcriptional reporter for cytochrome P450 oxidase, a phase I drug-detoxification enzyme.

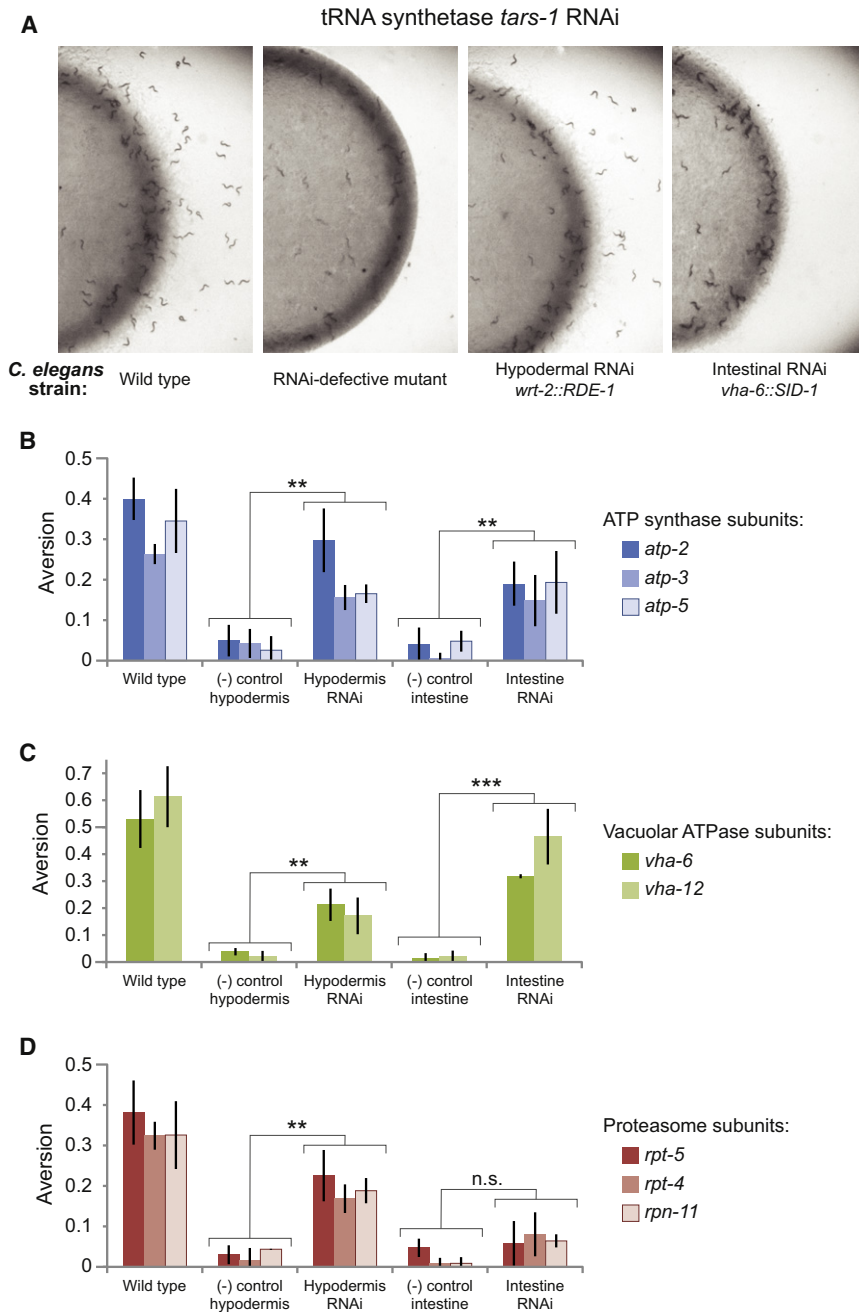
(C) RNAi of *eat-6* and *rpt-3* stimulates hypodermal and intestinal expression of a *gst-4::GFP* transcriptional reporter for glutathione S-transferase, a phase II detoxification enzyme.

See also Figure S2 and Table S2.

stimulated only in the tissue-specific RNAi strain where those genes are expressed (Figures S4A and S4B).

We tested whether tissue-restricted RNAi of essential functional classes identified in the aversion screen could stimulate

microbial aversion. We found that most essential gene classes could stimulate aversion from either hypodermis or intestine. Genes encoding three ATP synthase subunits (*atp-2*, *atp-3*, or *atp-5*) or two vacuolar ATPase subunits (*vha-6* and *vha-12*) could



induce aversion when inactivated in either the intestine or the hypodermis (Figures 4B and 4C). Although aversion phenotypes were milder than in wild-type animals, tissue-restricted RNAi animals exhibited significantly higher aversion than RNAi-defective strains. Not all essential functional classes tested stimulated aversive behavior from both tissues. For instance, hypodermal inactivation of proteasome subunits or tRNA synthetases stimulated aversion, whereas intestinal inactivations did not (Figures 4D, S4C, and S4D). Inactivation of protein translation by RNAi of ribosomal components in either tissue stimulated aversion (Figures S4C and S4D).

Figure 4. Neuroendocrine Control of Aversion Behavior

(A) Aversion is stimulated by inactivation of the threonyl tRNA synthetase *tars-1* in the hypodermis but not the intestine.

(B–D) Aversion is induced by hypodermis or intestine-restricted RNAi of (B) mitochondrial ATP synthase subunits and (C) vacuolar ATPase subunits and hypodermis-restricted RNAi of (D) proteasome subunits. Representative experiments $\pm$ standard deviation (SD) shown. * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$ by t test; n.s. not significant.

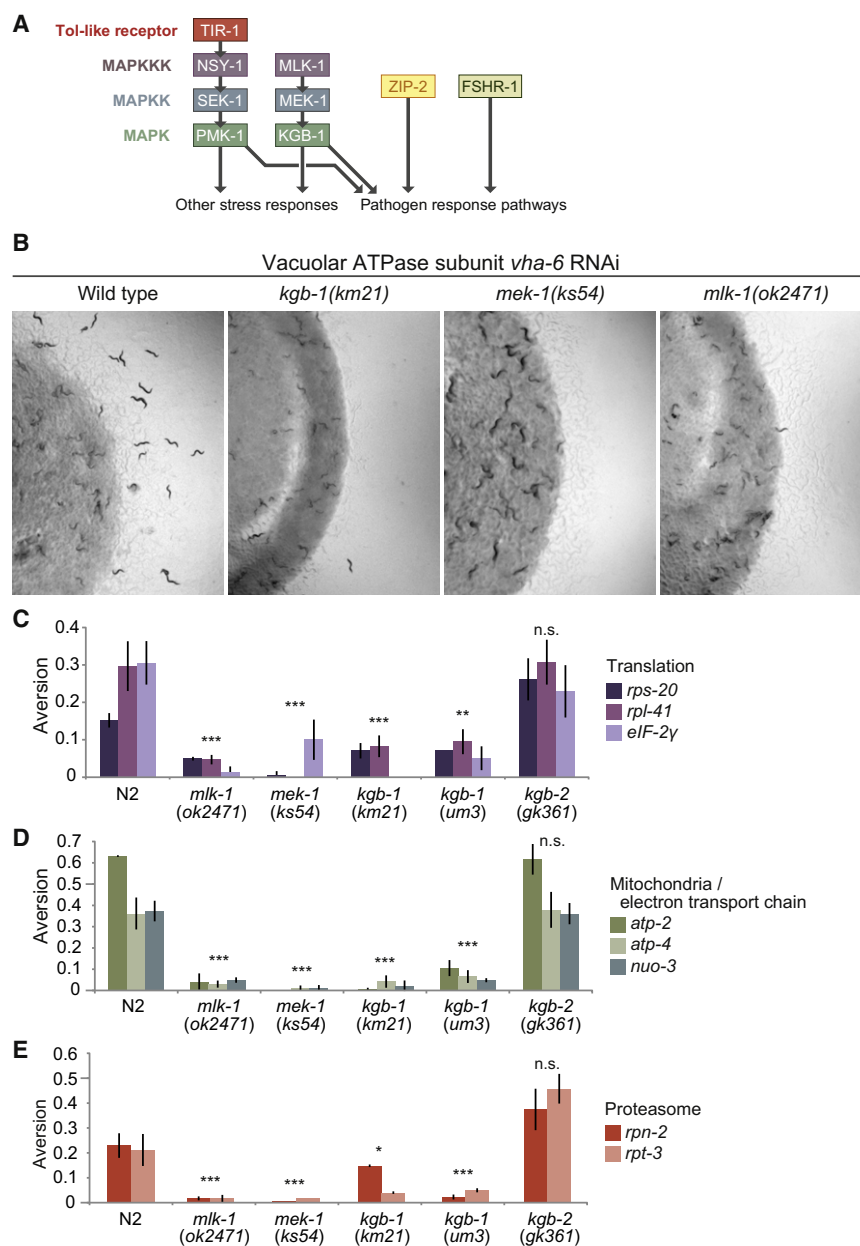
See also Figures S3 and S4 and Table S3.

Aversion Behavior Requires a Jnk-like MAP Kinase Pathway

Multiple pathways contribute to pathogen resistance in *C. elegans*. Immune responses to *P. aeruginosa* may be the most intensively characterized. Animals are protected from *P. aeruginosa* infection by at least three genetic pathways, represented by (1) FSHR-1, a GPCR specifically required in the intestine (Powell et al., 2009), (2) ZIP-2, a bZip family transcription factor controlling the induction of many pathogen-responsive genes (Estes et al., 2010), and (3) the NSY-1/SEK-1/PMK-1 p38 family MAP kinase cassette downstream of the TIR-1 Toll-like receptor necessary for innate immunity in many systems (Figure 5A). The p38 MAPK pathway was an especially attractive candidate because VHP-1, an inhibitory phosphatase of PMK-1 (Kim et al., 2004), stimulates microbial aversion when inactivated by RNAi (Table S1A). We examined mutants in these pathways for a role in aversion behavior. At least two mutant alleles were tested for all genes except *fshr-1* (Table S4). Mutants in none of these pathways attenuated microbial aversion behavior (Figures S5A and S5B and data not shown), although both *sek-1* mutant alleles exhibited a partial reduction in

aversion in response to RNAi of vacuolar ATPase, ATP synthase, and tRNA synthetase genes (Figure S5A and data not shown), without an accompanying requirement for *tir-1*, *nsy-1*, or *pmk-1*.

However, a Jnk-like MAP kinase pathway was required for the aversion response. MLK-1/MEK-1/KGB-1 comprise a MAP kinase cassette required for resistance to the ER toxin tunicamycin (Mizuno et al., 2008), transcriptional responses to pore-forming toxins (Kao et al., 2011), and *S. cerevisiae* infection (M. Yun and S. Politz, personal communication). The VHP-1 phosphatase antagonizes KGB-1 (as well as PMK-1) (Mizuno et al., 2004), so it seemed possible that *vhp-1* RNAi-induced



aversion behavior might be caused by hyperactivation of KGB-1. We also tested the *kgb-1* paralog *kgb-2*, for which no function has yet been reported. We found that the *mlk-1*, *mek-1*, and *kgb-1* genes were required for microbial aversion behavior in response to RNAi of all functional gene classes tested (Figures 5B–5E), although *kgb-1* was only partially required when tRNA synthetases or vacuolar ATPases were inactivated (Figure S5C). *kgb-2* was not required for aversion in any gene category. For *kgb-1*, *mek-1*, and *mlk-1* mutants, inactivation of ETC components (*atp-2*, *atp-4*, or *nuo-3*), protein translation factors (*rps-20*, *rpl-41*, or Y39G10AR.8, the eIF-2 γ homolog), or proteasomal subunits (*rpn-2* or *rpt-3*) did not elicit aversion significantly above basal levels (Figures 5C–5E). *kgb-1* and *sek-1* were each

Figure 5. The Jnk MAP Kinase Pathway Is Required for Aversion

(A) Four candidate pathogen and stress-response pathways in *C. elegans*.

(B) Aversion phenotypes for Jnk pathway mutants following inactivation of *vha-6* (vacuolar ATPase subunit).

(C–E) The Jnk pathway is required for aversion induced by RNAi of (C) translation components, (D) mitochondrial ETC components, and (E) proteasome subunits. Representative experiment $\pm$ SD shown. * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$ by t test. n.s. not significant.

See also Figure S5.

partially required for aversion induced by RNAi of vacuolar ATPases and tRNA synthetases (Figures S5A and S5C), suggesting that for some subtypes of cellular stress, these two kinases operate collectively to control the microbial aversion response.

Essential Gene RNAi and Pathogens Exhibit Similar Neurobiological Traits

C. elegans develops an avoidance of pathogenic bacteria that it initially finds attractive, possibly by pairing internal with chemosensory cues of infection (Schulenburg and Ewbank, 2007). Pathogen avoidance requires an intact chemosensory system (Styer et al., 2008) and operates via a serotonergic circuit (Zhang et al., 2005). Reduced ingestion of pathogenic microbes has also been reported (Schulenburg and Müller, 2004; Sicard et al., 2007). We examined the behavioral profiles of essential gene RNAi-treated animals to determine whether aversion behaviors stimulated by disruption of core cellular processes share similar neurological characteristics with pathogen-avoidance behavior.

We examined whether animals with prior experience of aversion-inducing “noxious” RNAi bacteria were capable of associating internal with chemosensory cues to guide future feeding decisions. RNAi and control RNAi (–) bacteria are the same *E. coli* K12 strain (HT115), differing only in whether they express dsRNA targeting a *C. elegans* gene. It seemed likely that expression of dsRNAs corresponding to a *C. elegans* gene would not significantly impact the repertoire of gustatory and olfactory signals produced by HT115. We predicted that if experience-dependent neural associations reinforce aversion behavior stimulated by disruption of cellular processes, animals raised on a noxious RNAi lawn should develop behavioral avoidance of both deleterious and benign HT115—but perhaps not against

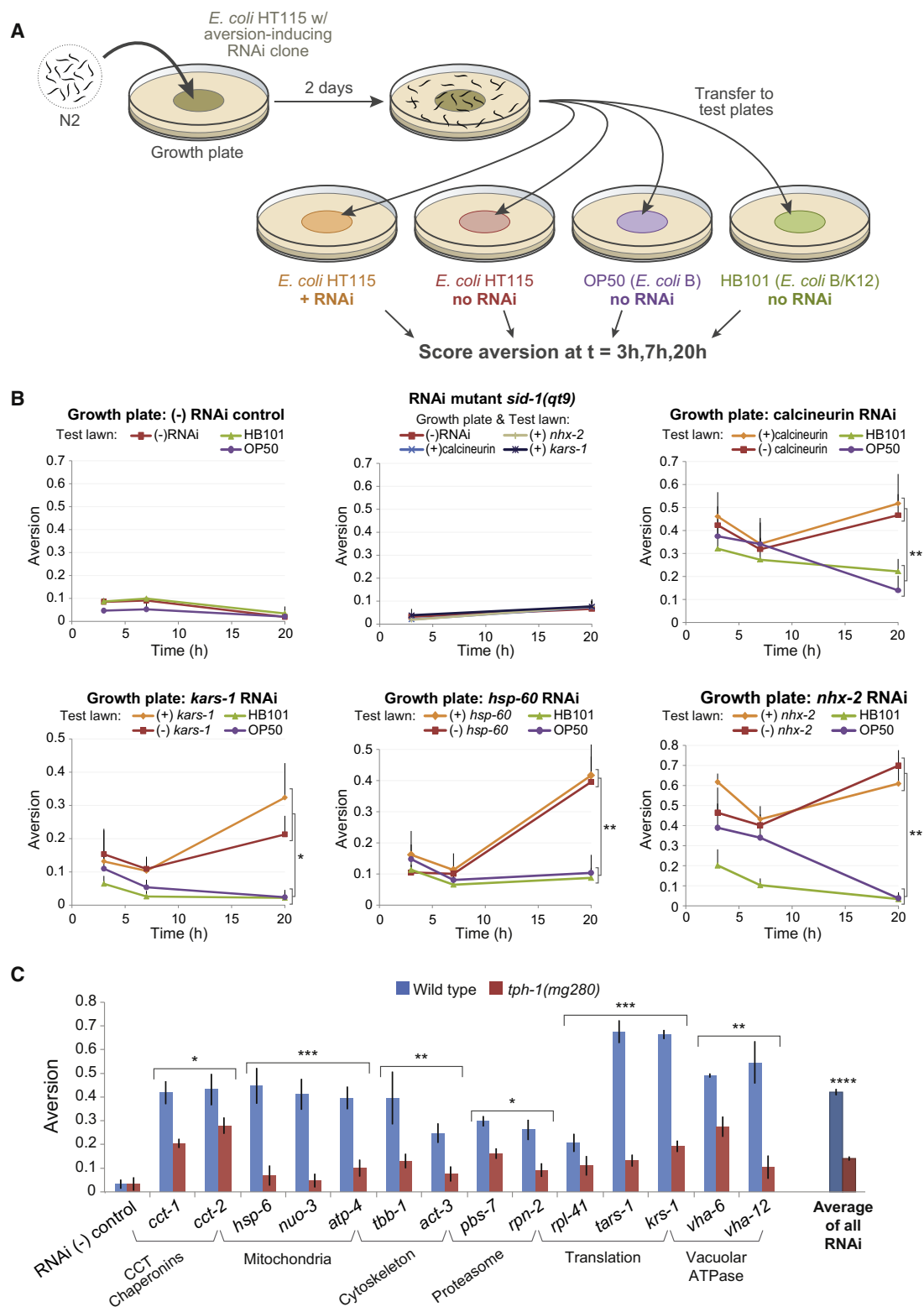


Figure 6. Learned Avoidance and Serotonergic Signaling Mediate Aversion Behavior

(A) Schematic outline of learning experiment.

(B) Representative aversion time courses for animals grown on RNAi bacteria inactivating the *kars-1* tRNA synthetase, *hsp-60* mitochondrial chaperone, *nhx-2* Na^+/K^+ potassium pump, or *Y71H2AL.1* calcineurin B. Control experiments: top left, animals grown on HT115 RNAi control and tested on HT115, HB101, or OP50

less-related *E. coli* or other bacterial strains. To test this prediction, animals were raised for 2 days on noxious RNAi “growth plates” and transferred to “test plates” containing one of four microbial options: (1) the same noxious RNAi bacteria upon which they were grown, (2) the benign RNA (–) control, (3) OP50 *E. coli*, or (4) HB101 *E. coli* (Figure 6A). Options 3 and 4 represent genetically distinct *E. coli* isolates from HT115. OP50 is an *E. coli* B strain, and HB101 is an *E. coli* B/K12 hybrid strain. Animals were assayed for aversion at 3, 7, and 20 hr after transfer to test plates. Four aversion-inducing gene inactivations were selected to disrupt a range of biological functions: *kars-1* (tRNA synthetase), *Y71H2AL.1* (calcineurin B subunit), *nhx-2* (Na⁺/H⁺ transporter), and *hsp-60* (mitochondrial chaperone). Animals raised on all four noxious RNAi lawns exhibited similar levels of microbial aversion for all four tester strains in the first 7 hr after transfer (Figure 6B). Between 7 and 20 hr, animals on the (–) control and noxious RNAi test plates exhibited sustained or increasing levels of aversion. In contrast, by 20 hr, animals transferred to OP50 or HB101 test plates had undergone a decline in aversion, often to near-basal levels for that bacterial strain (data not shown). Animals on the four test plates were of the same developmental stage and size, indicating that the reduction in lawn aversion on OP50 and HB101 plates was not attributable to attenuation of RNAi activity (Figure S6A). Animals raised on benign HT115 plates exhibited no aversion after transfer to tester plates (Figure 6B, top left panel). Similarly, RNAi-defective mutant animals displayed no aversion despite continuous growth on noxious RNAi bacteria (Figure 6B, top middle panel). We also tested several bacterial species that may represent more ecologically relevant food choices for *C. elegans* than laboratory *E. coli*. After 2 days of growth on *kars-1* RNAi plates, animals were transferred to test plates containing microbial species known to reside in soil environments: *B. subtilis* (a gift from R. Losick), *B. simplex*, and a *Comamonas* sp. soil isolate (gifts from L. Avery). At 2 and 5 hr, animals did not display significantly different aversion phenotypes on any tester plate, but by 14 and 22 hr, animals transferred to any of the three soil-associated species exhibited lower aversion than those transferred to *kars-1* RNAi and control HT115 plates (Figure S6B).

The observation that animals grown on noxious RNAi HT115 *E. coli* avoided benign HT115 *E. coli* suggested that neurosensory associations reinforce and sustain aversion behavior. We assayed aversion responsiveness in mutants defective for distinct aspects of neurosensory: chemosensation (*che-2(e1033)* and *che-3(e1124)*), mechanosensation (*mec-3(e1388)*, *mec-4(u45)*, and *mec-7(e1527)*), thermotaxis (*ttx-1(p767)* and *ttx-4(nj3)*), and aerotaxis (*gcy-35(ok769)*). We also tested the *tax-4(p678)* mutant, which inactivates a broadly expressed nucleotide-gated channel required for multiple neurosensory responses, including chemotaxis (Bargmann, 2006),

chemorepulsion, aerotaxis, and pathogen avoidance (Styer et al., 2008; Yook and Hodgkin, 2007). We expected that sensory-defective mutants might exhibit aversion phenotypes on benign bacteria, possibly due to deficits in chemical, mechanical, or oxygen-related aspects of food recognition. Similarly, we thought sensory mutants might fail to exhibit aversion behavior in response to noxious RNAi due to their inability to detect environmental cues necessary to form learned associations with internal signals. We measured baseline aversion levels of sensory mutants over a developmental time course on RNAi (–) control bacteria. Although several mutants exhibited slightly elevated basal aversion behavior, *che-2(1033)* was the only mutant exhibiting significantly higher aversion than wild-type by 48 hr ($0.18 \pm .05$ versus $0.02 \pm .01$, $p < 0.0001$, Student's t test) (Figure S6C). Because sensory mutant defects might be masked in an aversion assay by their naturally higher aversion rates, we plotted raw and baseline-subtracted aversion levels for all sensory mutants. Mutants were tested for aversion responses after 48 hr of growth on RNAi bacteria targeting the *vha-6*, *kars-1*, *cco-1*, or *elt-2* genes. Only *tax-4*, which is required for multiple sensory modalities, was strictly required for the aversion response in all four cases (Figures S6D–S6G). *che-2* and *che-3* mutants, which lack functional chemosensory neurons, exhibited no significant defect in response to aversion induced by *vha-6* or *kars-1* RNAi (Figures S6D and S6E) but were strongly defective for aversion behavior in response to RNAi of *cco-1* or *elt-2* (Figures S6F and S6G). None of the mutants impaired in mechanosensation, thermotaxis, or aerotaxis caused significant differences in aversion compared to wild-type.

We also tested how animals in “aversion” states respond to new sensory cues. Our learning data showing that growth on noxious RNAi bacteria caused animals to avoid all bacterial foods for up to 7 hr suggested that animals raised on noxious RNAi bacteria are not as initially attracted to positive stimuli as naive animals. To explore this possibility, we examined the stimulus responsiveness of RNAi-treated animals in a panel of olfactory, gustatory, and tactile assays that report on both attractive and repulsive behavioral responses. We first examined chemotaxis to the attractive odorant isoamyl alcohol in animals that were previously grown on noxious RNAi bacteria disrupting a range of cellular activities. Most gene inactivations did not interfere with chemoattraction, but several caused mild deficits (*qua-1*, *atp-3*, *ptc-1*, *pqn-47*, *vha-6* RNAi) (Figure S7A). Chemotaxis to the soluble attractant NH₄Cl was used to evaluate animal attraction to gustatory cues after growth on noxious RNAi bacteria (Figure S7B). Two gene inactivations (*cmd-1* and *kars-1* RNAi) showed significant defects in chemotaxis; several others were slightly less attracted to NH₄Cl than the RNAi control, but these reductions were not statistically significant. Five RNAi treatments were tested for avoidance of the volatile repellent, octanol, but no deficit in behavioral repulsion was

E. coli; top middle, *sid-1(qt9)* RNAi-defective animals were grown and tested on the same HT115+/– RNAi bacteria (e.g., grown and tested on *nhx-2* RNAi). Statistically significant differences between samples were only observed at the 20 hr time point. * $p < 0.01$; ** $p < 0.001$ by t test. For graphical clarity, bars show +SD only.

(C) Serotonin-deficient *tph-1(mg280)* animals exhibited a partial aversion defect (~35% of wild-type) in response to gene inactivations representing major functional classes. Representative experiments $\pm$ SD shown. * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$; **** $p < 10^{-19}$ by t test.

See also Figures S6 and S7.

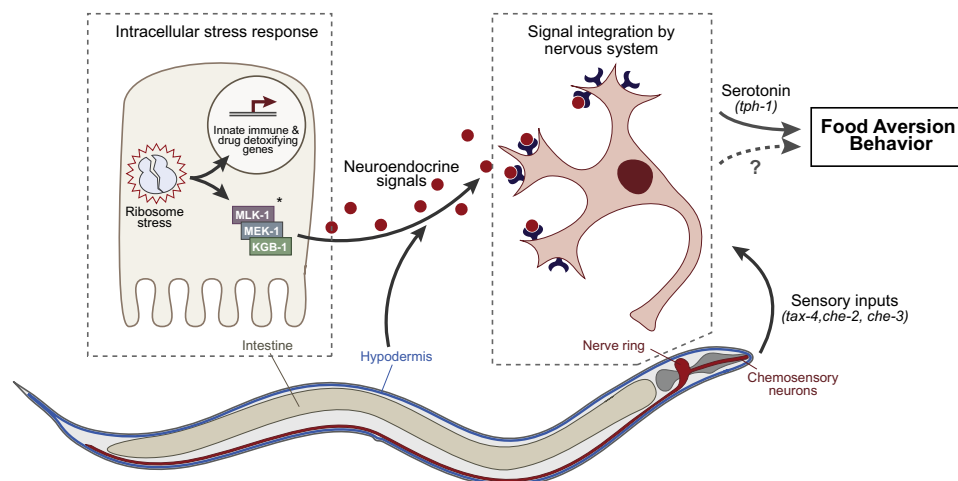


Figure 7. Model of Aversion Behavior

Surveillance pathways detect disruptions of core cellular processes in animal tissues (e.g., disruption of protein translation in the intestine, shown). An endocrine circuit involving the Jnk MAPK pathway relays stress signals to the nervous system, where they are integrated with multiple sensory inputs to control the microbial aversion response. *Although Jnk MAPK pathway components are depicted as acting in the intestine in the model diagram, the site(s) of action for this signaling cascade have not yet been determined. The p38 pathway MAPKK component SEK-1 (not shown in model) also contributes to microbial aversion in response to certain functional classes of gene inactivation, possibly in parallel with the KGB-1 MAP kinase.

detected (Figure S7C). A light touch assay was used to assess mechanosensory reactions, and none of the gene inactivations tested exhibited detectable defects in tactile avoidance (Figure S7D).

Pathogen-avoidance behavior requires serotonin signaling (Zhang et al., 2005). To determine whether noxious RNAi avoidance also engages a serotonergic circuit, we analyzed aversion responses in the serotonin biosynthetic mutant *tph-1(mg280)*. We found that serotonin-deficient animals exhibited a consistent ~65% reduction in aversion behavior relative to wild-type controls for every functional category tested (Figure 6C) ($p < 10^{-9}$, Student's *t* test).

In addition to locomotory avoidance of pathogens, some studies have reported a reduction in microbial ingestion (Yook and Hodgkin, 2007). We measured pharyngeal pumping rates for a panel of gene inactivations yielding fertile adults (so all samples are stage-matched with controls) (Figure S7E). Nine of ten RNAi treatments yielded a mild to strong reduction in pumping rate that was statistically significant ($p < 10^{-4}$, Student's *t* test with Bonferroni correction).

DISCUSSION

Food Avoidance Is a Behavioral Response to Disruption of Essential Cellular Activities

We have shown that *C. elegans* develops a behavioral aversion to otherwise nutritive, nonpathogenic *E. coli* when those bacteria produce dsRNA that inactivates *C. elegans* genes required for fundamental cellular activities—such as protein translation or oxidative respiration. The generality of aversion behavior as a response to functionally diverse gene inactivations allowed us to conduct an RNAi screen to identify the range of physiologic processes that may be subject to surveillance. From this screen,

we discovered that aversion was induced by disruption of protein translation, mitochondrial functions, the proteasome, vacuolar ATPases, the tubulin and actin cytoskeletons, mRNA processing, chromatin packaging, central metabolism, and the molting program.

Several cellular surveillance pathways are known and have been extensively studied, including the ER^{UPR} and DNA-damage and cytoskeleton checkpoints. Ribotoxic stress and mitochondrial stress response pathways have also been reported (Iordanov et al., 1997; Haynes and Ron, 2010). However, many of the cellular processes and molecules identified in our screen are not associated with previously reported surveillance pathways and may therefore reveal new cellular defense programs (Figure 7). Moreover, even those stress responses that are well-characterized should be re-examined for their potential roles in pathogen surveillance and defense.

Surveillance of Cellular Functions Provides a Mechanism for Toxin and Pathogen Detection

C. elegans are repelled by pathogenic bacteria or fungi, exhibiting both locomotory avoidance and reduced feeding (Schulenburg and Ewbank, 2007). Multiple tissues, including the intestine and hypodermis, are susceptible to pathogen invasion, and many microbial signals and host pathways are likely to be involved in mounting an effective immune response. Our results with innate immune and xenobiotic detoxification reporters suggest that inactivation of a single cellular process may be sufficient to mount both immune and detoxification responses. RNAi-mediated disruption of translation, mitochondrial functions, vacuolar ATPases, and basic metabolic pathways were all potent inducers of the *clec-60::GFP*, *irg-1::GFP*, and *F35H12.5::GFP* pathogen-responsive reporters—despite the absence of a pathogen or toxin in these experiments.

Pathogen-associated reporters were weakly stimulated by inactivation of proteasomal and secretory pathway components.

For the engagement of innate immune responses, mechanisms of pathogen recognition are thought to involve both direct sensation of pathogen-intrinsic signals (PAMPs) such as cell-wall components (Janeway, 1989; Jones and Dangl, 2006) and sensation of host-derived signals (DAMPs) caused by toxin or virulence factor-induced damage to host tissues (Matzinger, 1994). DAMPs are generally believed to be small molecules or cellular proteins released into the extracellular environment when host cells die. Our findings suggest that cellular surveillance pathways detect invading pathogens or xenobiotic compounds by sensing disruption to cellular processes (such as protein translation or oxidative respiration) caused by secreted toxins and virulence factors. We propose that these pathways are monitored by a cSADD system that mediates pathogen and xenobiotic detection and coupled responses. Upon detection of cellular disturbances, such surveillance mechanisms engage host defenses by stimulating xenobiotic detoxification genes to neutralize toxins and immune defenses to eradicate microbes, activating compensatory programs to restore function to pathways disabled by pathogens and toxins, and enabling escape from pathogens by locomotory avoidance and suppression of pathogen ingestion. One advantage of such a system is that it can detect decreases in core cellular functions before cell death, when DAMP signals are released. In contrast to natural pathogens, which transfer to hosts a suite of virulence factors and toxins with pathogenic functions that may be difficult to isolate, our system of induction of pathogen-response pathways by inactivation of individual host genes allows removal of many layers of evolved measures and countermeasures. This advantage will lend itself to dissection of individual surveillance pathways and the elucidation of corrective mechanisms for disruptions to particular processes.

In support of the interpretation of our findings outlined above, many of the essential cellular functions identified in our screen are known targets of microbial toxins. We found that supplementation of attractive *E. coli* food with microbial toxins (antimycin A, tunicamycin, and concanamycin A) targeting the same cellular processes identified in our screen (the mitochondrial ETC, the secretory pathway, and vacuolar ATPases) also triggers the aversion response. Analysis of the *gst-4::GFP* and *cyp-35B::GFP* detoxification reporters revealed that many essential gene inactivations engaged xenobiotic detoxification responses. These data suggest that toxins (and xenobiotics and medicinal drugs) are detected via the surveillance of the core cellular processes targeted by such poisons rather than by detection of the poison itself. Thus the detection of toxins and bacterial pathogens may employ common surveillance pathways and may couple to induction of cytochrome P450 and other detoxification systems as well as induction of antibacterial immunity functions. Because bacteria and fungi produce a large array of xenobiotic secondary metabolites, the products of polyketide synthase genes or nonribosomal peptide synthesis genes that constitute 5%–10% of many microbial genomes, it is plausible that most toxins are derived from bacteria, fungi, or plant matter on which the animal was feeding. Although toxins are diverse, they are biased toward the disruption of highly conserved cellular

machinery, such as ribosomal and mitochondrial components, perhaps to be pathogenic to the widest range of hosts. Significantly, the gene inactivations that most potently induce antibacterial behaviors and gene expression are among the most conserved *C. elegans* genes, suggesting that the “core proteome” is targeted by microbial toxins and virulence factors. Direct surveillance of these systems may enable the host to defend itself effectively from pathogens with which it has no evolutionary history. Other emerging studies support this cellular surveillance model of pathogen detection. Exposure of *C. elegans* to the *Pseudomonas aeruginosa* virulence factor ToxA, which disables translation elongation factor EF2 (one of the most highly conserved eukaryotic genes and an aversion hit in our screen), and hygromycin, which inhibits ribosome translocation, induces pathogen-response genes (McEwan et al., 2012), and inactivation of translation using RNAi or the translation inhibitor cycloheximide stimulates the pathogen-response genes (Dunbar et al., 2012). Both of these studies substantiate the previously reported identification of a “ribotoxic stress response,” stimulated by microbial toxins (e.g., Shiga toxin) that inhibit translation elongation in mammalian cells (Iordanov et al., 1997).

Control of Aversion through MAP Kinase and Endocrine Pathways

Cellular stress responses are coupled to endocrine pathways in *C. elegans*. Insulin signaling regulates dauer arrest in response to high population density or starvation. Disruption of the ETC in neurons activates the mitochondrial UPR in intestinal cells (Durieux et al., 2011), and neuroendocrine signals control proteotoxicity (Prahlaad and Morimoto, 2011) and heat shock responses (Prahlaad et al., 2008) in non-neuronal tissues. Using tissue-restricted RNAi, we have shown that the hypodermis and the intestine are each capable of stimulating the aversion response for many essential gene inactivations. Given that both tissues contact the external environment, they make attractive candidates as extraneuronal sensory organs for microbial toxins and virulence factors. However the expression patterns of some aversion genes suggest that additional tissues may be competent to trigger the aversion response, including the gonad, vulva, pharynx, body wall muscle, and excretory cell. Regardless of the range of tissues involved in sensation of cellular stressors, the finding that aversion can be triggered by disruption of core processes in non-neuronal tissues argues that neuroendocrine signals are relayed between peripheral tissues and the neurons that mediate aversion behavior.

We tested whether known pathogen-response pathways also mediate the aversion response. We found no evidence that three major pathways controlling *P. aeruginosa* resistance, the ZIP-2, FSHR-1, and p38 MAPK pathways, were involved in aversion behavior. However a Jnk-family MAP kinase pathway, comprised of MLK-1 (MAPKKK), MEK-1 (MAPKK), and KGB-1 (MAPK), involved in *C. elegans* responses to tunicamycin and pore-forming toxins was clearly required for microbial aversion behavior. All three kinases were essential for aversion in response to many gene inactivations, including mitochondrial ETC and proteasome genes. However, aversion induced by RNAi of translation factors and vacuolar ATPases exhibited

a shared dependency on KGB-1 and SEK-1, a p38 pathway component—suggesting that KGB-1 and SEK-1 function in parallel in certain cases. As mentioned above, the mammalian Jun and p38 MAP kinase pathways are activated when cells are exposed to toxins that inhibit elongation of protein translation (Iordanov et al., 1997), suggesting that *C. elegans* and mammals may respond to toxin-mediated disruption of translation and other cellular functions via a common signaling network.

Similarities of Pathogen-Avoidance Behavior and RNAi-Induced Food Aversion

Although aversion can be stimulated by cues emerging from non-neuronal tissues, our analysis of neurosensory mutants demonstrated that aversion behavior also requires an intact neurosensory system. *tax-4*, which encodes a cyclic nucleotide-gated channel, is required for many sensory responses and for pathogen-avoidance behavior. *tax-4* was fully required for induction of aversion by inactivation of essential cellular pathways. However, mutants in individual sensory modalities (touch, heat, oxygen sensation) had little or no effect on aversion responses. *che-2* and *che-3* were required for aversion induced by *cco-1* and *elt-2* RNAi but not *vha-6* or *kars-1* RNAi, suggesting that chemosensation may be required for aversion behavior in specific instances. Together these findings show that aversion relies on the neurosensory system but is likely redundantly reinforced by multiple sensory modalities. We also found that, like pathogen-avoidance behavior, noxious RNAi-induced aversion behavior exhibits a significant requirement for serotonin. On average, ~65% of the aversion response relies on serotonergic circuit(s), indicating that other forms of neurotransmission must provide the residual aversion observed in serotonin mutants.

The ability of *C. elegans* to integrate sensory with undefined internal cues may be the basis for learned pathogen-avoidance behavior. We have shown that animals reared on aversion-inducing RNAi *E. coli* (the HT115 strain) develop and maintain an avoidance of HT115, even when shifted to an HT115 lawn that does not produce noxious dsRNAs, consistent with pathogen-avoidance association studies. Previous experience with noxious RNAi bacteria not only causes animals to avoid the HT115 strain, it causes them to avoid unrelated benign bacterial species at levels comparable to the noxious RNAi strain for up to 7 hr. After 7 hr, aversion behavior on non-HT115 bacteria subsided so that by 20 hr, aversion was barely detectable. This result suggests that once entrained on noxious bacteria, animals may adopt an “aversive state” that is resistant to modulation by new sensory cues, even attractive ones. This proposition is supported by our preliminary findings that preconditioning on noxious RNAi bacteria can cause reduced chemotaxis to attractive olfactory or gustatory cues but had no observable effect on behavioral responses to aversive stimuli.

A Surveillance System for Pathogens and Xenobiotics

Our screen for microbial aversion behavior has shown that a wide range of critical cellular processes are monitored by surveillance pathways, many of which are not associated with known cellular surveillance systems. Many of the cellular processes identified are targets of toxins produced by bacteria likely to be found in the natural ecosystem of any animal or plant.

Given that aversion-associated genes are among the most conserved across eukaryotic phylogeny, host surveillance pathways are likely to be shared among eukaryotes. Consistent with this view, we found that cellular surveillance signals are transduced through a conserved JNK-family MAP kinase cascade to produce the aversion response. We would also suggest that cellular surveillance systems detect xenobiotics and medicinal drugs via their inhibition of core cellular pathways and may likewise couple (through JNK or related MAP kinase pathways) to behavioral modifications and detoxification responses. Because inactivation of a cellular pathway can act as a signal of bacterial infection, toxicity caused by medicinal drugs may in some cases be due to inappropriate induction of antibacterial behaviors or engagement of innate immune defenses. For example, if nausea represents the human analog of “microbial aversion behavior,” then the nausea produced by chemotherapeutic treatments may indicate the misapplication of a normally adaptive response to the ingestion of pathogen-tainted foods.

The physiological diversity of virulence factors and bacterial toxins represents the billion year arms race with eukaryotic surveillance and detoxification systems: as the host evolves surveillance, detoxification, and immunity functions, bacteria will evolve countermeasures that may neutralize the surveillance pathways required for immune defense. There may be corresponding complexity and genetic variation in the surveillance pathways marshaled by eukaryotes, perhaps based on their varied toxin and pathogen histories. Such variation in humans may underlie distinct responses to drugs, therapeutic or addictive, or to normally benign chemicals or foods. The molecular components of these surveillance pathways, such as the conserved JNK kinase cascade, may constitute new drug targets to alleviate the nausea and appetite suppression caused by many medicines or the toxic side effects that derail many drugs in development. Inactivation of these surveillance pathways may also suppress aberrant induction of xenobiotic and pathogen-response pathways in a variety of diseases.

EXPERIMENTAL PROCEDURES

Microbial Avoidance Assays

Saturated overnight cultures were concentrated 30–40× for HT115 RNAi bacteria or ~20× for OP50 bacteria (used in drug assays) in S Basal; 50 µl aliquots were dropped to 6-well RNAi plates and left on the benchtop overnight. For toxin avoidance, toxins were dropped to lawns ~1 hr prior to use. Synchronized L1s (for RNAi) or L4/A stage (for drugs) were dropped directly onto lawns, and wells were scored for aversion ($A = N_{\text{off lawn}}/N_{\text{total}}$) at desired time points. Each RNAi or drug treatment was conducted in triplicate in each experiment and was repeated in two or more independent trials.

Drug concentrations used can be found in the [Extended Experimental Procedures](#).

Neurosensory and Feeding Assays

Chemotaxis, mechanosensation, and pharyngeal pumping experiments were performed using standard procedures (Hart, 2006).

Statistical Methods

The Student's *t* test and chi-square tests were used to validate behavioral results as stated in the figure legends. DAVID bioinformatic analysis (<http://david.abcc.ncifcrf.gov/home.jsp>) was used to identify enriched gene classes from the RNAi screen. Multiple comparison correction was made for several experiments.

See the [Extended Experimental Procedures](#) for detailed descriptions of behavioral assays, animal imaging, tissue-restricted RNAi experiments, and statistical methods. See [Table S4](#) for an inventory of *C. elegans* and bacterial strains used in this study.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Extended Experimental Procedures, four tables, and three movies and can be found with this article online at [doi:10.1016/j.cell.2012.02.050](https://doi.org/10.1016/j.cell.2012.02.050).

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